

Quantification of Neutral Lipids

Using Nile Red Staining.

Nile Red, a lipophilic fluorescent dye, selectively stains neutral lipids, allowing lipid content to be assessed through fluorescence intensity. Using a 96 well assay, yeast cells are stained with Nile Red and fluorescence will be measured at excitation 485 nm and emission 535 nm. Yeast cells can also be imaged using fluorescence microscopy

Reagents

✓	Name	Quantity	Location/Notes
☐	DMSO	1.25mL	
☐	PBS (1X)	1.25mL	
☐	Nile Red (2 mg/mL)	60 µL	
☐	Acetone	1940 µL	
☐	Yeast cells	250 µL	See protocol for preparation
☐	10 % (v/v) formalin (only for imaging)	100 µL	
☐	SlowFade Antifade Mountant (only for imaging)	~20 µL	

Equipment

✓	Name	Quantity	Location/Notes
☐	Black, clear bottom 96 well plate		
☐	Plate Reader		

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| ☐ Centrifuge | | |
| ☐ Fluorescence microscopy (only for imaging) | | |
| ☐ Slides and coverslip (only for imaging) | | |

Instructions

96-Well Plate Quantification Assay

[Prepare the reagents]

- Prepare 60 $\mu\text{g}/\text{mL}$ Nile Red working solution by diluting 60 μL of 2 mg/mL Nile Red stock (in acetone) into a final volume of 2 mL acetone.
 - a. 60 μL Nile Red stock (2 mg/mL) + 1940 μL acetone
- Prepare DMSO:PBS (1:1, v/v) by mixing equal volumes of 1.25 mL DMSO and 1.25 mL 1X PBS.

[Cell preparation]

- Harvest cells by centrifugation for 2 min at $492 \times g$.
- Wash pellets twice with 1X PBS.
- Resuspend cells in 1X PBS at an OD (595nm). Final resuspension should be in total volume of at least 1 mL .
- Record OD 595 nm. Blank is PBS. Adjust so that OD 595 nm is around 1.0.

[Quantification]

- Using a black, clear bottom 96 well plate, per well add 25 μL DMSO:PBS (1:1) to each well required for the assay.
- Add 250 μL of sterile 1 $\times$ PBS into at least 6 of the DMSO:PBS containing wells of row A. Add 25 μL of the prepared Nile Red solution as well.
 - a. These are the blank wells
- 250 μL of yeast suspension, 25 μL of the prepared Nile Red solution
 - a. Final volume is 300 μL , Nile Red = 5 $\mu\text{g}/\text{mL}$
- Plate each sample as triplicates
- Read the plate immediately after addition of the Nile red dye to the wells for the required number of cycles

- Plate reader settings: wavelengths excitation 485 nm and emission 535 nm on top 50 % mirror, using the optimal gain and orbital shake for 10 s before reading

Extra notes:

- Vortex mixing of the 2 mg/mL Nile red stock in acetone before preparation of the working solution helped with homogeneity
- Make the 60 µg/mL working Nile red solution fresh for each experiment
- When washing cells in preparation for the Nile red assay with 1× PBS, allow the cells to remain in 1× PBS for 1–2 min between washes improves the sensitivity of the assay.

Fluorescence Microscopy

[Prepare the reagents]

1. Prepare 60 µg/mL Nile Red working solution by diluting 60 µL of 2 mg/mL Nile Red stock (in acetone) into a final volume of 2 mL acetone.
 - a. 60 µL Nile Red stock (2 mg/mL) + 1940 µL acetone
2. Prepare DMSO:PBS (1:1, v/v) by mixing equal volumes of DMSO and 1X PBS.

[Staining and Imaging]

3. Transfer 250 µL yeast culture into sterile 1.5 mL tubes.
4. Add 25 µL DMSO:PBS (1:1) and 25 µL Nile Red working solution (60 µg/mL).
5. Gently mix and incubate 5 min in the dark at room temperature.
6. Centrifuge 2 min at 492 × g, remove supernatant.
7. Wash pellet twice using 1X PBS, centrifuge sample after each wash.
8. Resuspend pelleted cells with 100 µL 10 % (v/v) formalin and fix for 15 minutes in the dark at room temperature.
9. Centrifuge for 2 min at 492 × g and wash twice with 1X PBS, centrifuge sample after each wash.
10. Resuspend the final cell pellet in a small volume of SlowFade (~20 µL).
11. Pipette 5 µL onto a slide, add the coverslip. Can seal the edges of the coverslip with nail varnish to prevent drying out.
12. Image using a fluorescence microscope
 - a. Excitation: ~485 nm
 - b. Emission: ~535–575 nm

https://link.springer.com/protocol/10.1007/978-1-4939-6788-9_16#Sec10

References

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